SIMATS ENGINEERING

SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES

CO3-AT2

ASSESSMENT TOOL 2 – DATA INTERPRETATION

Detailed Solutions — Questions 16 to 20

Submitted for

DSA0404 & DSA0405 – Fundamentals of Data Science

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This report presents complete, examiner-ready solutions to Questions 16–20 of CO3-AT2: Data Interpretation. Each question follows a uniform six-part structure — Question, Given Data, Steps to Solve, Entire Solution (full working), Final Answer, and Conclusion — so that the reasoning process, not just the final number, is fully visible.

KEY FORMULAS USED IN THIS REPORT

The table below consolidates every statistical formula applied across Questions 16–20, so the underlying logic of each solution can be traced back to its formal definition.

Concept	Formula
Z-score (standardisation)	$Z = (X - \mu) / \sigma$
Standard Error of the Mean	$SE = \sigma / \sqrt{n}$ (or $s / \sqrt{n}$ for a sample)
Confidence Interval (mean)	$\bar{x} \pm Z_{\alpha/2} \times SE$
Interquartile Range	$IQR = Q3 - Q1$
1.5×IQR Outlier Fences	$Lower = Q1 - 1.5(IQR)$; $Upper = Q3 + 1.5(IQR)$
Covariance	$Cov(X,Y) = Sum[(X_i - \bar{X})(Y_i - \bar{Y})] / n$
Pearson Correlation Coefficient	$r = Cov(X,Y) / (SD_x \times SD_y)$
Pearson's Skewness (median-based)	$Sk = 3(Mean - Median) / SD$
Population Variance	$\sigma^2 = \Sigma(X_i - \mu)^2 / n$
Population Standard Deviation	$\sigma = \sqrt{\sigma^2}$
Empirical Rule	$\mu \pm 1\sigma \approx 68\%$, $\mu \pm 2\sigma \approx 95\%$, $\mu \pm 3\sigma \approx 99.7\%$

How to read this report: every question below reproduces the exact question text, lists the data actually given, walks through the solving strategy in plain language before any arithmetic is shown, then performs the full numeric working, states the final answer in one highlighted line, and closes with a short conceptual takeaway. This mirrors how a full-mark answer should be structured in an exam script — reasoning first, numbers second, interpretation last.

Question 1 | Continuous Distribution – Area Interpretation

1. Question

A continuous random variable X is normally distributed with mean $\mu = 100$ and standard deviation $\sigma = 15$. (a) What proportion of values lie between 85 and 130? (b) A value of 145 is observed – how many standard deviations away is it, and would you flag it as a potential outlier under the empirical rule?

2. Given

Parameter	Value
Distribution	Normal (Gaussian)
Mean (μ)	100
Standard Deviation (σ)	15
Interval of interest	85 to 130
Observed value	145

3. Steps to Solve

Step 1: Convert each raw boundary (85, 130, 145) into a standard Z-score using $Z = (X - \mu) / \sigma$, since areas under a normal curve are only readily available in the standard normal (Z) table.
Step 2: Look up (or recall) the cumulative probabilities $\Phi(Z)$ for each Z-score from the standard normal distribution.
Step 3: For part (a), subtract the lower cumulative probability from the upper cumulative probability to get the area (proportion) between 85 and 130.
Step 4: For part (b), compute the Z-score of 145 and compare its magnitude against the empirical (68-95-99.7) rule thresholds of $\pm1\sigma$, $\pm2\sigma$ and $\pm3\sigma$ to decide whether it should be flagged as an outlier.

4. Entire Solution

Part (a): Proportion of values between 85 and 130

```
Z(85) = (85 - 100) / 15 = -15 / 15 = -1.00
Z(130) = (130 - 100) / 15 = 30 / 15 = +2.00
```

So the interval [85, 130] corresponds to exactly $[\mu - 1\sigma, \mu + 2\sigma]$ on the standard normal scale — it is not symmetric about the mean (1 SD on the left, 2 SD on the right).

From the standard normal table:

```
Φ(2.00) = P(Z ≤ 2.00) = 0.9772
Φ(-1.00) = P(Z ≤ -1.00) = 0.1587
```

The required area is the difference between these two cumulative probabilities:

```
P(-1.00 < Z < 2.00) = Φ(2.00) - Φ(-1.00)
= 0.9772 - 0.1587
= 0.8185
```

Therefore approximately **81.85%** of all values are expected to fall between 85 and 130. This can be cross-checked intuitively: the empirical rule says ~68% of data lies within ±1σ (85 to 115) and ~95% lies within ±2σ (70 to 130). Our interval (85 to 130) covers the entire right half up to 130 (2σ above) plus the left half only up to 85 (1σ below), so a value between the 68% and 95% figures — 81.85% — is exactly what we would expect.

Part (b): How unusual is the value 145?

```
Z(145) = (145 - 100) / 15 = 45 / 15 = 3.00
```

The observed value of 145 is exactly **3 standard deviations above the mean**. Under the empirical (68-95-99.7) rule for a normal distribution:

Range	Expected Coverage	145 falls here?
μ ± 1σ (85 to 115)	≈68%	No
μ ± 2σ (70 to 130)	≈95%	No
μ ± 3σ (55 to 145)	≈99.7%	Exactly at the boundary

Since 145 sits precisely on the ±3σ boundary, it lies in the extreme outer 0.3% tail of the distribution (only about 0.15% of values are expected to exceed it on the upper side). Values at or beyond ±3σ are conventionally treated as rare/extreme observations.

5. Answer

✓ ANSWER: (a) P(85 < X < 130) ≈ 81.85%. (b) Z = 3.00 → 145 is exactly 3 standard deviations above the mean; under the empirical rule this places it in the extreme <0.3% tail, so YES, it should be flagged as a potential outlier.

6. Conclusion

Conclusion: *Converting raw scores to Z-scores lets us read exact tail probabilities directly off the standard normal distribution rather than relying only on the coarse 68-95-99.7 rule. A Z-score of 3.0 is a widely used statistical threshold for flagging extreme values, since under normality less than 0.3% of observations are expected to lie beyond it — making 145 a strong candidate for further investigation rather than a routine reading.*

Common Mistakes to Avoid

• Forgetting that the interval 85 to 130 is *asymmetric* around the mean (–1σ to +2σ) and wrongly applying the flat “95% within 2σ” rule to it directly.
• Mixing up Φ(Z) (cumulative probability up to Z) with the point density of the normal curve at Z.
• Reporting the Z-score of 145 without connecting it back to the empirical rule thresholds to justify the outlier flag.

Question 2 | Sampling Bias vs Sampling Error

1. Question

A survey estimating average household expenditure samples only online panel respondents. The sample mean is Rs. 18,000 vs a known population mean of Rs. 14,500 (from census data). (a) Is this discrepancy better explained by sampling error or sampling bias? (b) Would increasing the sample size fix this problem? Justify.

2. Given

Parameter	Value
Sampling frame	Online panel respondents only
Sample mean expenditure	Rs. 18,000
Population mean (census, true value)	Rs. 14,500
Discrepancy	Rs. 3,500 (≈24.1% overestimate)

3. Steps to Solve

Step 1: Distinguish the two possible sources of discrepancy conceptually — *sampling error* is random, unavoidable fluctuation that shrinks as sample size grows; *sampling bias* is a systematic distortion caused by a flawed sampling method that does NOT shrink with more data.
Step 2: Examine how the sample was actually drawn — here, only online panel respondents were surveyed, which is a non-random, non-representative subset of the population.
Step 3: Reason about who is systematically excluded (offline / non-internet households) and how that group likely differs in expenditure from the online panel.
Step 4: Decide whether enlarging n would correct a random fluctuation (it would) or a structural flaw in who gets sampled (it would not), and justify the final verdict.

4. Entire Solution

(a) Sampling error or sampling bias?

Sampling error is the natural, random variation between a sample statistic and the true population parameter that occurs purely because we observed a subset rather than the whole population — it is unbiased in direction (equally likely to over- or under-estimate) and its expected magnitude is captured by the standard error, $SE = \sigma/\sqrt{n}$. Crucially, pure sampling error narrows as n grows and, on average, is centred at zero.

Sampling bias, in contrast, is a systematic error introduced by *how* the sample was collected. Here the survey draws exclusively from an online panel. Online panels systematically **exclude** households without reliable internet access or the digital literacy/inclination to join a panel — groups that, in most populations, skew toward lower-income and lower-expenditure households. The remaining online-panel population is therefore skewed toward wealthier, more urban, more tech-engaged respondents whose household expenditure is genuinely higher than the population average.

The gap observed (Rs. 18,000 vs Rs. 14,500, a 24% overstatement) is large, one-directional, and consistent with a known structural flaw in the sampling frame (coverage/selection bias) — not the kind of small, symmetric wobble that random sampling error produces. This points clearly to **sampling bias**, specifically a form of **coverage/selection bias** (sometimes called non-response or frame bias when a specific sub-population is systematically excluded from ever being selected).

(b) Would increasing sample size fix it?

No. Increasing the sample size only reduces the *standard error* of the estimate ($SE = \sigma/\sqrt{n}$ shrinks as n grows), which tightens the estimate around whatever mean the sampling process is actually centred on. But if every additional respondent is still drawn exclusively from the same biased online panel, the sampling process remains centred on the wrong (inflated) population — the online-panel sub-population, not the true general population. A bigger biased sample simply gives a more *precise* estimate of the wrong number: the 95% confidence interval would narrow around Rs. 18,000 rather than moving toward Rs. 14,500.

The only way to fix a bias problem is to fix the sampling *design* itself — for example, using a probability-based sampling frame that includes offline households (e.g. random-digit-dialling, door-to-door, postal, or mixed-mode surveys), applying post-stratification weighting to reweight the online sample so it matches known census demographics, or supplementing the online panel with a representative offline booster sample.

5. Answer

✓ ANSWER: (a) The discrepancy is explained by sampling bias (specifically coverage/selection bias from restricting the frame to online panel respondents), not random sampling error. (b) No — increasing sample size would only make the biased estimate more precise, not more accurate; it would not close the gap with the true population mean of Rs. 14,500.

6. Conclusion

Conclusion: *This case illustrates a fundamental distinction in survey statistics: precision (controlled by sample size) and accuracy (controlled by sampling design) are independent properties. No amount of additional data can compensate for a flawed sampling frame — only a redesigned, more representative sampling method can correct systematic bias.*

Common Mistakes to Avoid

• Assuming any gap between a sample statistic and a population parameter is automatically “sampling error” without examining how the sample was actually collected.
• Believing a larger sample size always “fixes” accuracy problems — it only fixes precision (random variation), never systematic bias.
• Failing to name the specific type of bias (coverage/selection bias here) and instead giving only a vague answer such as “the sample was bad.”

Question 3 | Covariance Sign vs Correlation Strength

1. Question

Two variable pairs both show positive covariance: Pair 1 has Cov = 850, Pair 2 has Cov = 12. Explain why you cannot conclude Pair 1 has a "stronger relationship" than Pair 2 without additional information, and state what that missing information is.

2. Given

Pair	Covariance	Sign
Pair 1	850	Positive
Pair 2	12	Positive

3. Steps to Solve

Step 1: Recall the mathematical definition of covariance and note that it carries the product of the original units of X and Y — it is **not** a unit-free quantity.
Step 2: Recognise that the magnitude of covariance depends heavily on the scale/variance of the underlying variables, so a bigger number does not automatically mean a "tighter" linear relationship.
Step 3: Recall that Pearson's correlation coefficient $r = Cov(X,Y) / (\sigma_X \sigma_Y)$ normalises covariance by the standard deviations of both variables, producing a bounded, unit-free measure of relationship strength.
Step 4: Conclude what missing information (the standard deviations of X and Y for each pair) is required before any comparison of "strength" can be made.

4. Entire Solution

$$Cov(X,Y) = \Sigma[(X_i - Xbar)(Y_i - Ybar)] / n$$

Notice that covariance retains the original measurement units of both variables (e.g. if X is in kilograms and Y is in rupees, Cov(X,Y) is in "kilogram-rupees"). This means its numeric size is directly inflated or deflated by however large or small the variances of X and Y happen to be — regardless of how tightly the points actually cluster around a line.

For example, Pair 1 (Cov = 850) could involve variables measured on large scales (say, company revenue in lakhs and marketing spend in lakhs) where even a loose, scattered relationship produces a large covariance purely because the variances of X and Y are large. Pair 2 (Cov = 12) could involve small-scale variables (say, hours studied and quiz score out of 10) where even a near-perfect, very tight linear relationship produces a small covariance purely because the variances are small.

To make a fair strength comparison, covariance must be standardised into Pearson's correlation coefficient:

$$r = Cov(X,Y) / (\sigma_X \times \sigma_Y)$$

Because r divides out the standard deviations of both variables, it is dimensionless and always bounded between -1 and +1, making it directly comparable across completely different pairs of variables measured in completely different units. Only after computing r for each pair can we legitimately say which relationship is "tighter"/"stronger".

Illustrative check: Suppose Pair 1 has $\sigma_X=50$, $\sigma_Y=40$ → $r = 850/(50×40) = 850/2000 = 0.425$ (moderate). Suppose Pair 2 has $\sigma_X=4$, $\sigma_Y=3$ → $r = 12/(4×3) = 12/12 = 1.0$ (perfect). Here Pair 2, despite the much smaller raw covariance, actually has the far stronger (indeed perfect) linear relationship — the exact opposite of what comparing raw covariance values alone would have suggested.

5. Answer

✓ **ANSWER: You cannot conclude Pair 1 is "stronger" because covariance is scale-dependent (its size reflects the variances/units of X and Y, not just how tightly the points cluster around a line). The missing information required is the standard deviations (σ_X, σ_Y) of both variables in each pair, so that covariance can be normalised into Pearson's correlation coefficient, $r = Cov/(\sigma_X \sigma_Y)$, before any true strength comparison is made**

6. Conclusion

Conclusion: *Covariance tells you only the direction of a linear relationship (positive, negative, or none) — never its strength in isolation. Comparing raw covariance values across different variable pairs is a common statistical pitfall; correlation, not covariance, is the correct standardised measure for judging and comparing relationship strength.*

Common Mistakes to Avoid

- Treating covariance as if it were already a "strength score" comparable across different variable pairs.
- Forgetting that covariance's units are the product of X's and Y's original units, so its magnitude is scale-dependent.
- Overlooking that correlation, unlike covariance, is always bounded in [-1, +1], which is precisely what makes it comparable across pairs.

Question 4 | Estimation – Confidence Interval Reasoning

1. Question

A sample of n = 100 items has mean = 250 and SD = 40. Construct a 95% confidence interval for the population mean. Then explain, in interpretation (not formula) terms, what this interval does and does NOT tell you about the true population mean.

2. Given

Parameter	Value
Sample size (n)	100
Sample mean (xbar)	250
Sample standard deviation (s)	40
Confidence level	95% (Z = 1.96)

3. Steps to Solve

Step 1: Compute the standard error of the mean, SE = s/√n, since n = 100 is large enough for the sampling distribution of the mean to be approximately normal (Central Limit Theorem).
Step 2: Identify the critical Z-value for a 95% confidence level (Z = 1.96, since 95% of the standard normal distribution lies within ±1.96 of the mean).
Step 3: Compute the margin of error, ME = Z × SE.
Step 4: Construct the interval as sample mean ± margin of error.
Step 5: Translate the numeric interval into a correct statistical interpretation, being careful to distinguish "confidence about the procedure" from an (incorrect) "probability statement about the fixed true mean."

4. Entire Solution

```
SE = s / √n = 40 / √100 = 40 / 10 = 4
Margin of Error (ME) = Z × SE = 1.96 × 4 = 7.84
95% CI = xbar ± ME = 250 ± 7.84
Lower limit = 250 - 7.84 = 242.16
Upper limit = 250 + 7.84 = 257.84
```

Quantity	Value
Standard Error (SE)	4.00
Margin of Error (95%)	±7.84
95% Confidence Interval	(242.16, 257.84)

What the interval DOES tell you:

It tells you that the procedure used to build this interval — "take a sample of 100, compute the mean, add/subtract 1.96 standard errors" — is a procedure that, if repeated across many independent samples, would produce an interval containing the true population mean about 95% of the time. It also gives a sensible, data-driven *range of plausible values* for the true population mean given the evidence in this one sample, and conveys how precise the estimate is (a range of about ±7.84 around 250).

What the interval does NOT tell you:

It does **not** mean there is a 95% probability that the true population mean falls within 242.16 to 257.84 for *this specific interval*. The true population mean is a fixed (though unknown) constant — it either lies inside this particular interval or it does not; there is no randomness left in it once the sample is drawn. The 95% refers to the long-run reliability of the *method* across repeated sampling, not to a probability attached to this one already-computed interval. It also does not tell you anything about where any individual data point lies, nor does it guarantee this particular interval is "correct" — roughly 1 in 20 such intervals, by chance, will miss the true mean entirely.

5. Answer

✓ ANSWER: SE = 4, Margin of Error = ±7.84, so the 95% Confidence Interval for the population mean is (242.16, 257.84). Interpretation: we are 95% confident — in the long-run-procedure sense — that this range brackets the true population mean, NOT that there is a 95% probability the fixed true mean lies in this specific interval.

6. Conclusion

Conclusion: *A confidence interval quantifies the reliability of an estimation procedure, not the probability of a fixed parameter. Correctly interpreting it as "95% of intervals built this way would capture the true mean" rather than "95% chance the mean is in this interval" is essential for sound statistical communication and avoids one of the most common misinterpretations in applied statistics.*

Common Mistakes to Avoid

• Saying "there is a 95% probability the true mean lies between 242.16 and 257.84" — this misapplies probability to a fixed, already-realised interval.
• Using the Z-value for a different confidence level (e.g. 1.645 for 90%, 2.576 for 99%) by mistake instead of 1.96 for 95%.
• Forgetting to divide the standard deviation by √n when computing the standard error, and instead using the raw SD as the margin-of-error base.

Question 5 | Integrated Case Study – Retail Daily Sales

1. Question

A retail company's daily sales data (in Rs. '000) for 15 days is: 120, 125, 118, 130, 122, 128, 500, 119, 124, 121, 126, 123, 117, 129, 125. (a) Identify the outlier and justify using both the IQR method and z-score method. (b) Compute mean and median with and without the outlier — compare the sensitivity of each. (c) Compute skewness with the outlier included — classify the distribution shape. (d) Recommend whether the company should remove, cap, or investigate the outlier before reporting monthly performance, and justify your recommendation using the statistical evidence from (a)–(c).

2. Given

Raw data (n = 15, Rs. '000): 120, 125, 118, 130, 122, 128, 500, 119, 124, 121, 126, 123, 117, 129, 125.
Sorted data: 117, 118, 119, 120, 121, 122, 123, 124, 125, 125, 126, 128, 129, 130, 500.

3. Steps to Solve

Step 1: Sort the data and compute the five-number summary (Q1, median, Q3) needed for the 1.5×IQR outlier rule.
Step 2: Apply the 1.5×IQR rule to obtain lower/upper fences and check which values fall outside them.
Step 3: Compute the overall mean and standard deviation, then the Z-score of the suspect value, and compare it against the conventional |Z| > 3 threshold.
Step 4: Recompute the mean and median after removing the flagged outlier, and compare how much each measure shifted (sensitivity comparison).
Step 5: Compute skewness (median-based Pearson coefficient) using the full dataset (outlier included) and classify the shape.
Step 6: Synthesise (a)-(c) into a single, evidence-based business recommendation about how to treat the value of 500 before reporting monthly performance.

4. Entire Solution

(a) Identifying the outlier

IQR Method:

Lower half (positions 1-7): 117, 118, 119, 120, 121, 122, 123 → Q1 = median = 120
Upper half (positions 9-15): 125, 125, 126, 128, 129, 130, 500 → Q3 = median = 128

```
IQR = Q3 - Q1 = 128 - 120 = 8
Lower fence = Q1 - 1.5 × IQR = 120 - 12 = 108
Upper fence = Q3 + 1.5 × IQR = 128 + 12 = 140
```

Since 500 > 140, the value **500 is confirmed as an outlier** by the IQR method (every other value in the dataset lies safely inside [108, 140]).

Z-score Method:

```
Σx = 120+125+118+130+122+128+500+119+124+121+126+123+117+129+125 = 2227
Mean (xbar) = 2227 / 15 = 148.47
Population Variance = Σ(x-xbar)² / n ≈ 132620.74 / 15 ≈ 8841.38
Standard Deviation (σ) = √8841.38 ≈ 94.03
```

```
Z(500) = (500 - 148.47) / 94.03 = 351.53 / 94.03 ≈ +3.74
```

A Z-score of approximately **+3.74** is well beyond the conventional |Z| > 3 outlier threshold, independently confirming what the IQR method already showed.

(b) Mean and median — with vs without the outlier

Measure	With Outlier (n=15)	Without Outlier (n=14)	Change
Mean	148.47	123.36	–25.11 (≈16.9%)
Median	124.00	123.50	–0.50 (≈0.4%)

Removing the single outlier changes the mean from 148.47 to 123.36 — a drop of 25.11 points (about 16.9% of the original value). The median, by contrast, barely moves at all, from 124 to 123.5 — a change of only 0.5 (about 0.4%). This starkly demonstrates that the **mean is highly sensitive** to extreme values (it is pulled directly by every data point, including the outlier), while the **median is highly robust/resistant** to outliers (it depends only on rank/position in the ordered data, not the extreme magnitude of any single value).

(c) Skewness with the outlier included

Using Pearson's median-based coefficient of skewness:

```
Sk = 3 × (Mean - Median) / SD
Sk = 3 × (148.47 - 124.00) / 94.03
Sk = 3 × 24.47 / 94.03
Sk = 73.40 / 94.03 ≈ +0.78
```

The coefficient is positive and moderately large (≈0.78), and Mean (148.47) > Median (124.00). This confirms the distribution is **positively (right) skewed** — the bulk of daily sales cluster tightly in the 117-130 range, while the single extreme value of 500 stretches a long tail out to the right, dragging the mean well above the median.

(d) Recommendation: remove, cap, or investigate?

The evidence from (a)-(c) should drive the decision, not a reflexive rule. Both the IQR method (500 > upper fence of 140) and the Z-score method (Z ≈ 3.74) independently and strongly confirm 500 is a genuine statistical outlier, and part (c) shows it single-handedly distorts the mean into a positively skewed, unrepresentative figure while leaving the median almost untouched.

Given this, the recommended action is to **investigate the value first**, rather than silently deleting or capping it outright:

• If investigation reveals a data-entry error or a system glitch (e.g. a duplicated/misplaced decimal, or a refund misrecorded as sales), the value should be **corrected or removed** before reporting, since it does not reflect real business activity.
• If investigation confirms it is a genuine spike (e.g. a festival sale, bulk B2B order, or promotional event), it should be **retained but reported with context** — for example, reporting both the median (a stable, typical-day figure of Rs. 123.5k) and a footnoted mean, or using Winsorization (capping at the upper fence of Rs. 140k) specifically for trend/average calculations, so the summary statistics are not misleadingly inflated while the underlying event is not erased from the record.

• In either case, the **median (Rs. 124k) is the safer headline statistic** for describing "typical" daily performance in this monthly report, since it is unaffected by whichever way the investigation resolves.

Blind removal risks silently discarding a real business event (loss of signal); blind inclusion in the plain mean risks giving management a badly inflated impression of average daily sales (Rs. 148k vs the true typical level of Rs. 123-124k). Investigation is the only option that protects both statistical accuracy and business truthfulness.

5. Answer

✓ ANSWER: (a) 500 is the outlier — confirmed by both IQR (500 > upper fence 140) and Z-score (Z ≈ +3.74). (b) Mean drops from 148.47 to 123.36 (≈16.9% change) while median barely moves from 124 to 123.5 (≈0.4% change) — mean is far more outlier-sensitive than median. (c) Skewness ≈ +0.78 → positively (right) skewed distribution. (d) Investigate the value of 500 before deciding to remove or cap it; report the median as the headline "typical day" figure.

6. Conclusion

Conclusion: *This integrated case ties together every concept in the assignment: outlier detection (IQR and Z-score), the differential sensitivity of mean versus median, and the link between outliers and skewness. The overarching lesson for real-world reporting is that a single extreme value can silently distort headline averages, so robust statistics (median, IQR) should always be reported alongside the mean, and any flagged outlier should be investigated for its real-world cause before it is mechanically deleted, capped, or accepted at face value.*

Common Mistakes to Avoid

• Using only one outlier-detection method (IQR OR Z-score) when the question explicitly asks for both, and not cross-checking that they agree.

• Automatically deleting the outlier without considering whether it might represent a genuine, business-relevant event.

• Reporting only the mean for "average daily sales" in the presence of a confirmed outlier, instead of pairing it with the more robust median.

SUMMARY OF FINDINGS

Q.No	Core Statistical Idea	Key Result
16	Normal distribution & empirical rule	$P(85<X<130)\approx81.85\%$; $Z(145)=3.0 \rightarrow$ outlier
17	Sampling bias vs sampling error	Bias (frame flaw); larger n will NOT fix it
18	Covariance vs correlation	Raw covariance is scale-dependent; use r to compare
19	Confidence interval interpretation	95% CI = (242.16, 257.84); procedure-based meaning
20	Integrated outlier & skewness analysis	500 is an outlier; distribution is right-skewed

Across Questions 16–20, the recurring theme is that summary statistics must always be interpreted together, never in isolation. A mean without a check on skewness or outliers can mislead; a covariance without its paired standard deviations cannot be compared; a confidence interval without correct procedural framing can be misquoted; and a sampling discrepancy without examining the collection method can be wrongly blamed on “bad luck” rather than “bad design.” Sound data interpretation depends on pairing every number with the reasoning that explains what it can, and cannot, tell us.

End of Report — CO3-AT2, Questions 16 to 20